

A Developer's Guide to Large Language Models and Prompt Engineering

Large language models are trained on massive datasets, enabling them to generate intelligent content. Prompt engineering is the art of creating efficient prompts to direct these models to get the intended results. Both must focus on responsible AI, so that the content that is generated is unbiased, respectful and in line with moral values.



Generative AI is a subset of deep learning. It uses artificial neural networks, which can process both labelled and unlabelled data using supervised, unsupervised, and semi-supervised methods. Large language models (LLMs) are also a subset of deep learning. Deep learning models or machine learning models, in general, can be classified into two types — discriminative and generative.

The discriminative model learns the conditional probability distribution or $p(Y|X)$, and it classifies as a dog and not a cat. The generative model captures the joint probability $p(X, Y)$ or just $p(X)$ if there are no labels, predicts the conditional probability that this is a dog, and can then generate a

picture of a dog (Figure 1). In essence, generative models can create new data instances while discriminative models discriminate between different data instances.

Figure 2 explains the distinction between generative AI and traditional AI.

LLMs are types of foundation models trained on massive datasets, enabling them to understand and generate natural language and other forms of content to perform various tasks. Microsoft has embraced interfaces such as Open AI's ChatGPT-3 and GPT-4, which make LLMs readily accessible

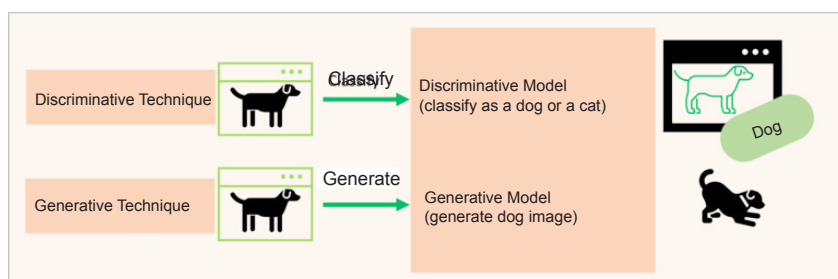


Figure 1: Predictions